

Combining time series microbiome data and genome-scale metabolic models predicts the metabolic underpinnings of time-lagged competitive/cooperative relationships between microbes

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Background

- Although the composition of the human gut microbiome has been extensively characterized in healthy and diseased states, fundamental ecological relationships among most gut bacteria remain unclear.
- Bacterial co-occurrence networks often capture indirect interactions, like microbes with similar oxygen needs cohabitating, rather than direct interactions like resource sharing or competition.
- Understanding these dynamics and their drivers is vital for developing tailored microbial communities to exclude or promote specific taxa.

Hypothesis: Direct, metabolically-driven interactions between bacterial taxa in the gut can be identified by time-lagged correlation modeling combined with genome- and community-scale metabolic modeling.

In this study, we used a one-day time lagged correlation strategy to identify pairs of bacteria whose abundances correlate more than expected by chance. To further understand these interactions and our ability to predict them, we used MICOM community-scale metabolic modeling parameters optimized to our dataset. The combination of these approaches has identified interacting taxa of interest and will allow us to further study the metabolic underpinnings of these interactions.

Gut Microbiome Time Series Data

- 16S rRNA from stool swabs
- 100 days including travel from Colorado, USA to Dhaka, Bangladesh
- Diarrheal illness experienced during travel (red vertical shading)
- Fairly consistent family level relative abundances

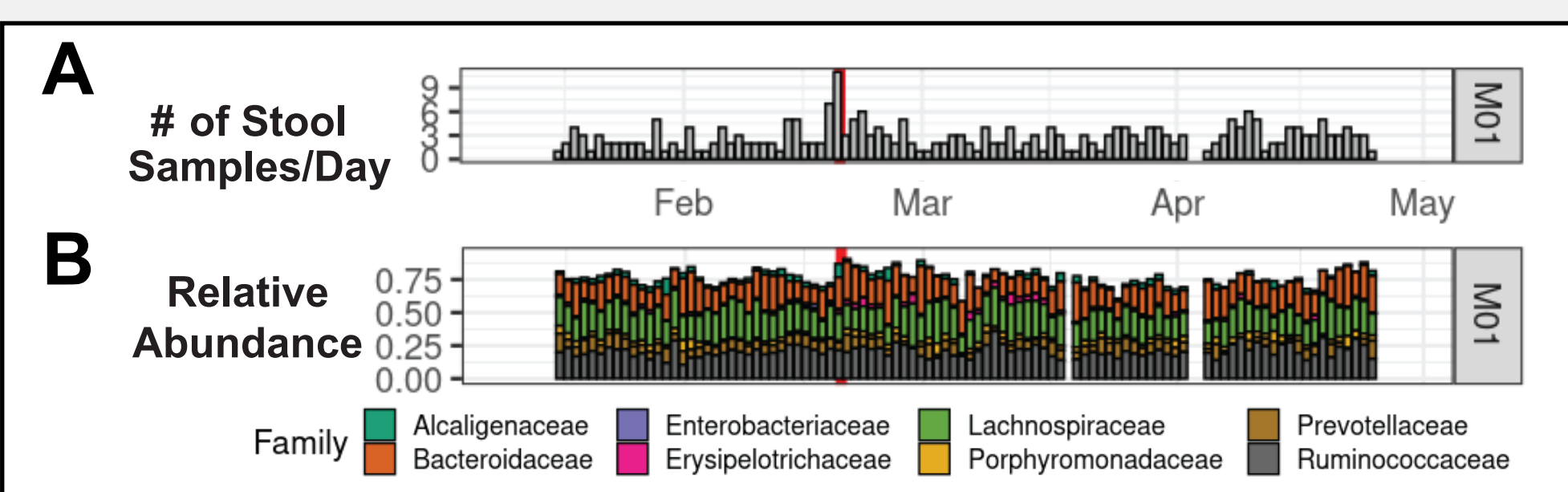
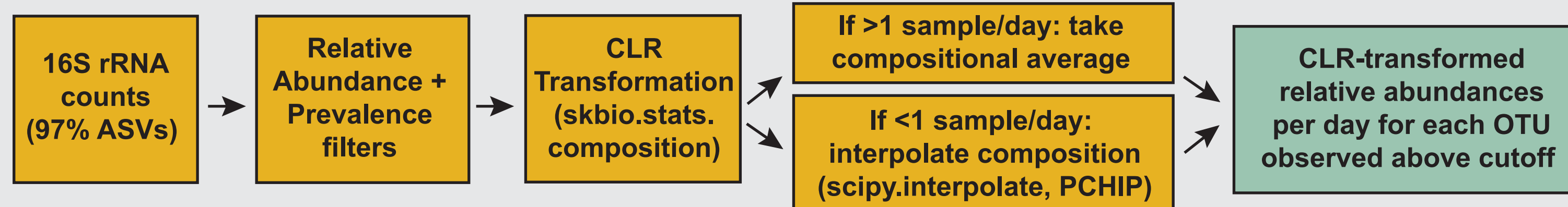


Figure 1. Characteristics of fecal data obtained from a single subject (M01) over time. (A) Number of stool samples collected per day, showing a peak during diarrheal illness. (B) Taxonomic composition (family level) over time.

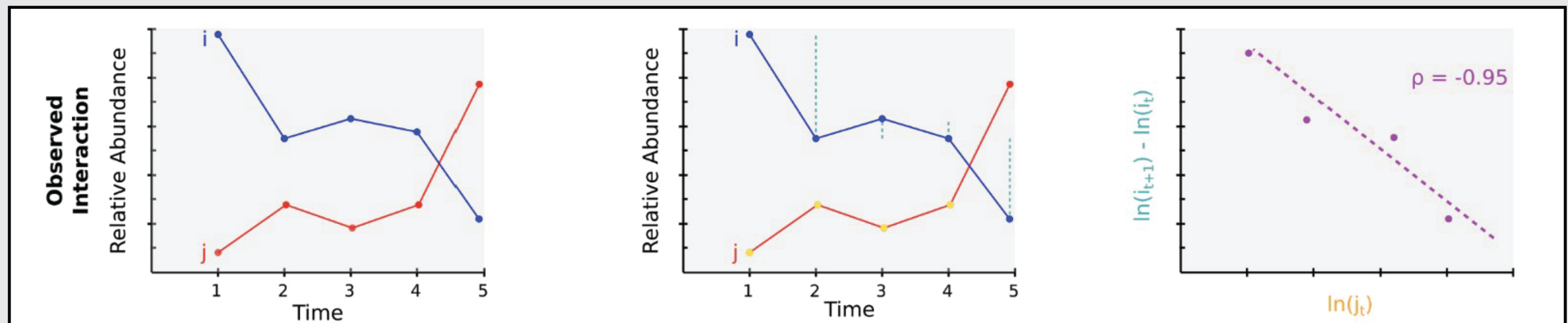
Time-lagged Correlation (TLC) Modeling



Time-lagged correlation basics as applied by Trosvik and de Muinck [1]

Does the abundance of one taxonomic unit (OTUj, red line) at one time (t) correlate with changes in another taxon's abundance (OTUi, blue line) at the next timepoint (t+1)?

Spearman rho (ρ)
Positive → Cooperation
Negative → Competition



Determination of statistically significant interactions

Previous shuffling/permutation-based methods employed for microbiome TLC significance failed to account for autocorrelation inherent to time series microbiome data.

Generation of time series surrogate data via *Iterative Amplitude Adjusted Fourier Transformation (IAAFT)* accounts for autocorrelation by preserving power spectrum & amplitude distribution of the original series [2].

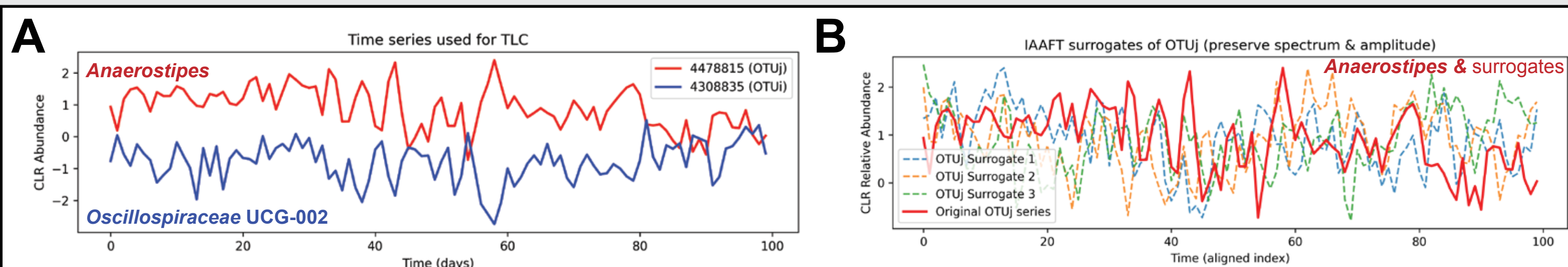


Figure 2. Example of time series surrogate generation and testing.

(A) Time series of OTUj (red, *Anaerostipes*) vs. OTUi (blue, *Oscillospiraceae* UCG-002) abundances in M01. (B) The original OTUj time series is shown with 3 IAAFT surrogates generated by pynicorn timeseries.surrogates.

(C) Spearman rho value distribution between 100,000 OTUj surrogates and the true OTUi time series. The dotted line indicates where the observed OTUj-OTUi correlation falls on the distribution. Here, *Anaerostipes* and *Oscillospiraceae* UCG-002 group have a significant positive correlation (p=0.0001).

This process is repeated for all possible interaction partners and p-values are corrected by Benjamini-Hochberg FDR to determine significant interactions (p.adj < 0.05).

Phylogenetic relatedness of negatively interacting taxa points to metabolic competition

Previous work from our group has shown that closely related taxa tend to share more similar metabolic input requirement sets (left).

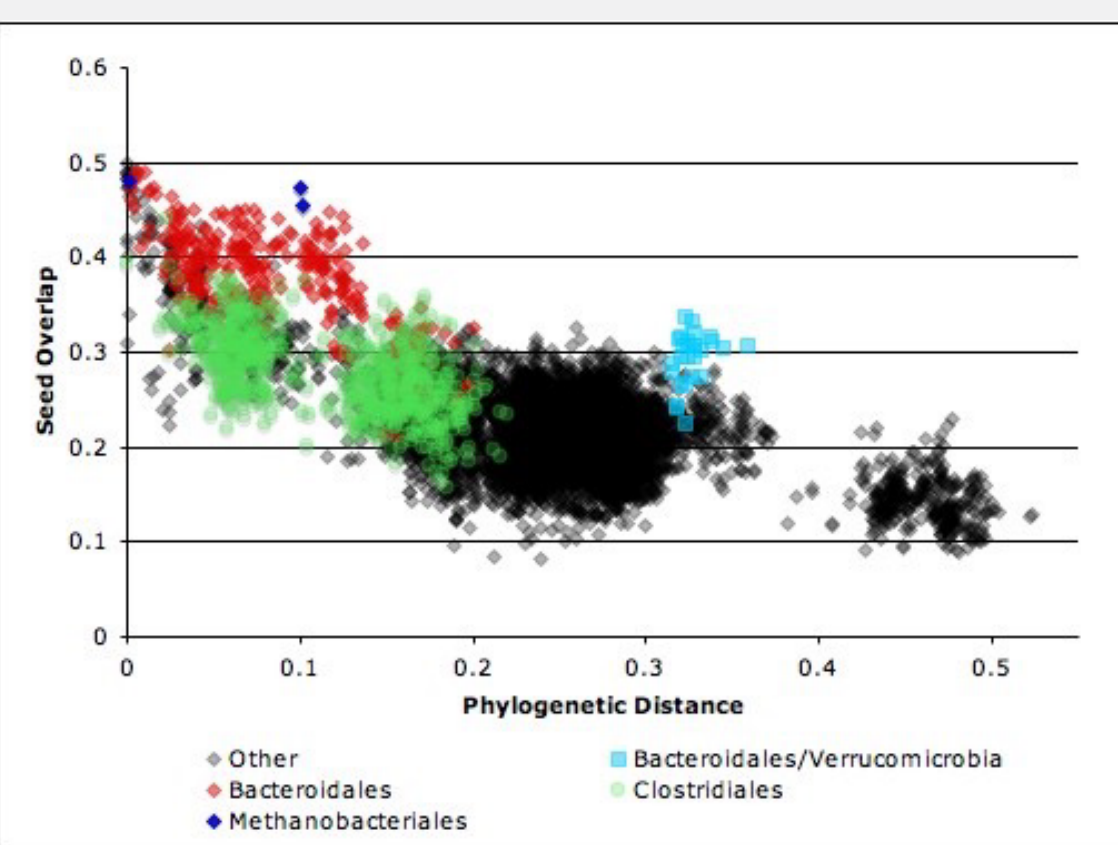
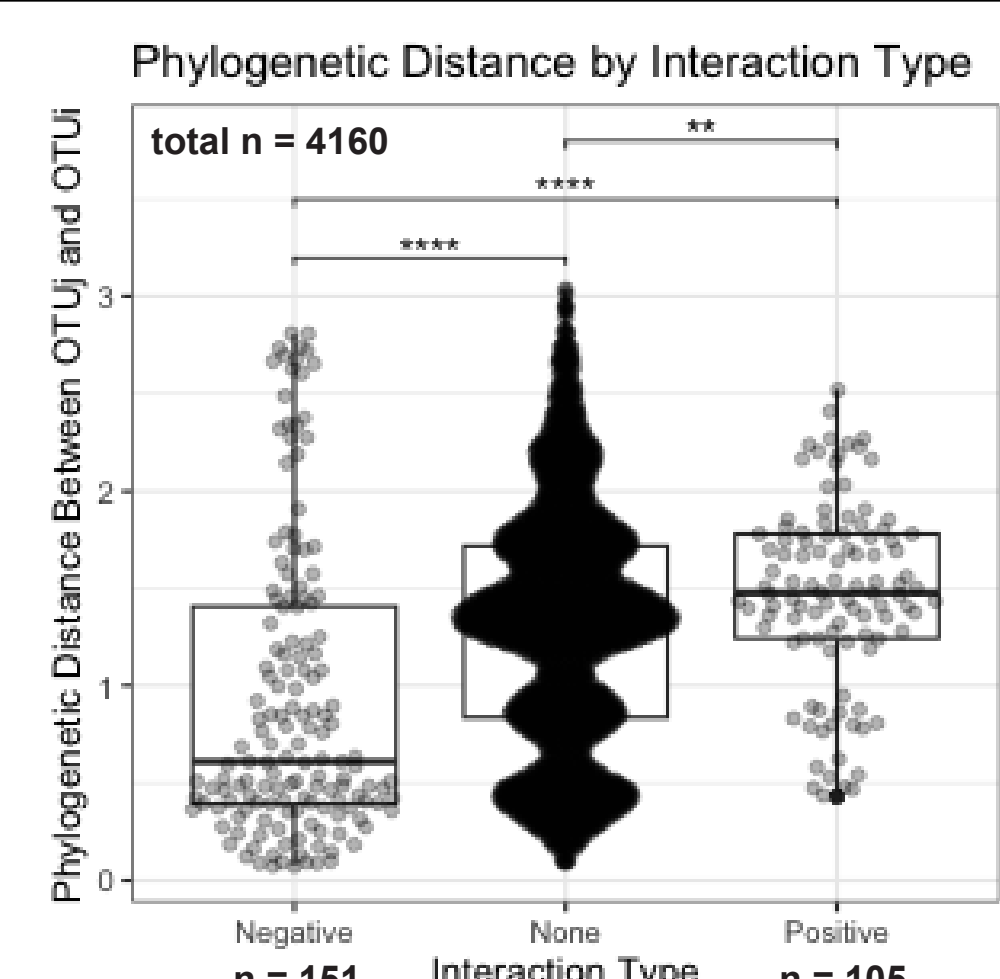
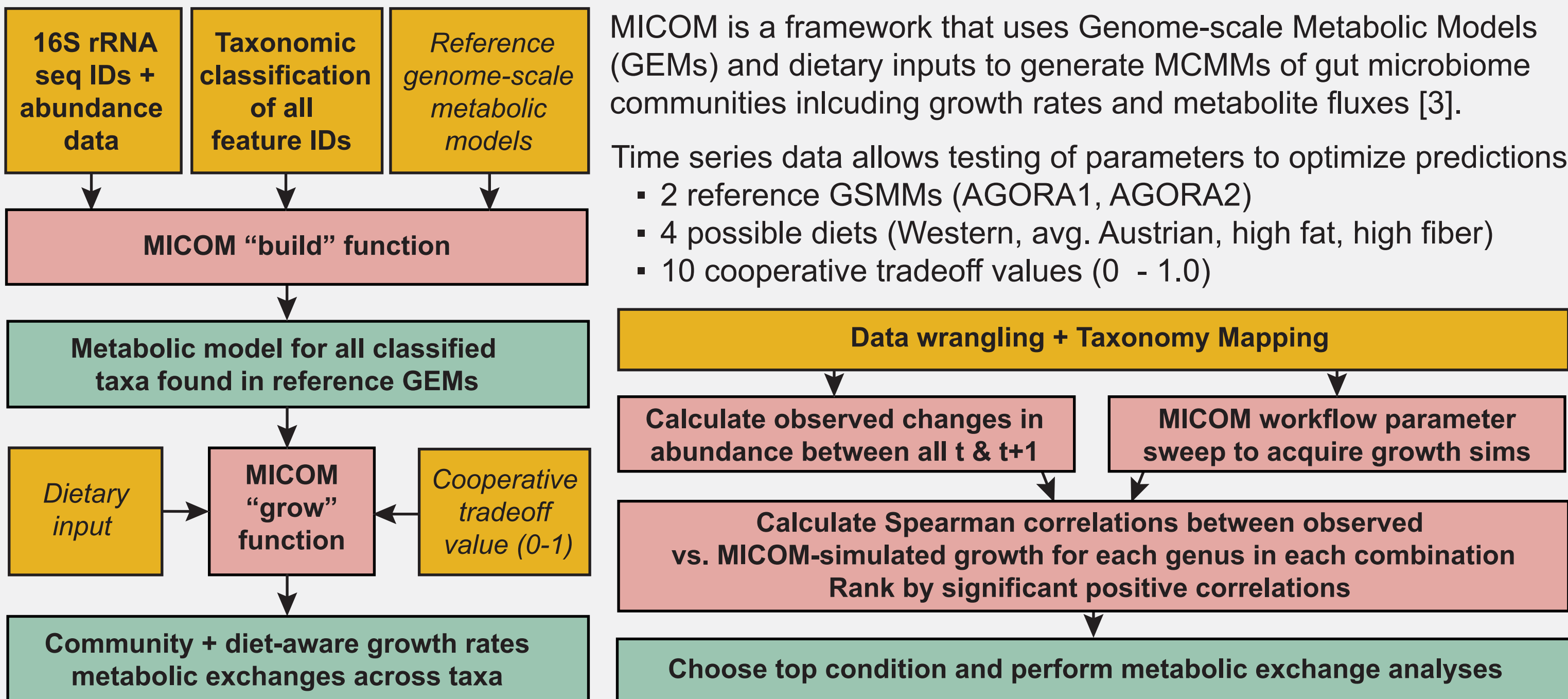


Figure 3 (right): Phylogenetic distance between interacting OTUs. All taxa were placed into the SILVA reference tree, and the patristic distance between each pair was calculated for significant negative, null, and significant positive time-lagged interactions. Negative interactions occurred between more closely related taxa and positive between less related taxa (**** p < 0.0001, ** p < 0.01, Wilcoxon with BH).



Stronger competition among related taxa implies that shared metabolic demands shape microbial community structure. We next used **microbial community-scale metabolic models (MCMs)** to evaluate whether the observed time-lagged interactions correspond to predicted metabolic flux relationships.

Microbial Community-scale Metabolic Modeling



Under optimized parameters, MICOM-predicted growth rates were significantly correlated with observed abundance changes in 58% of taxa

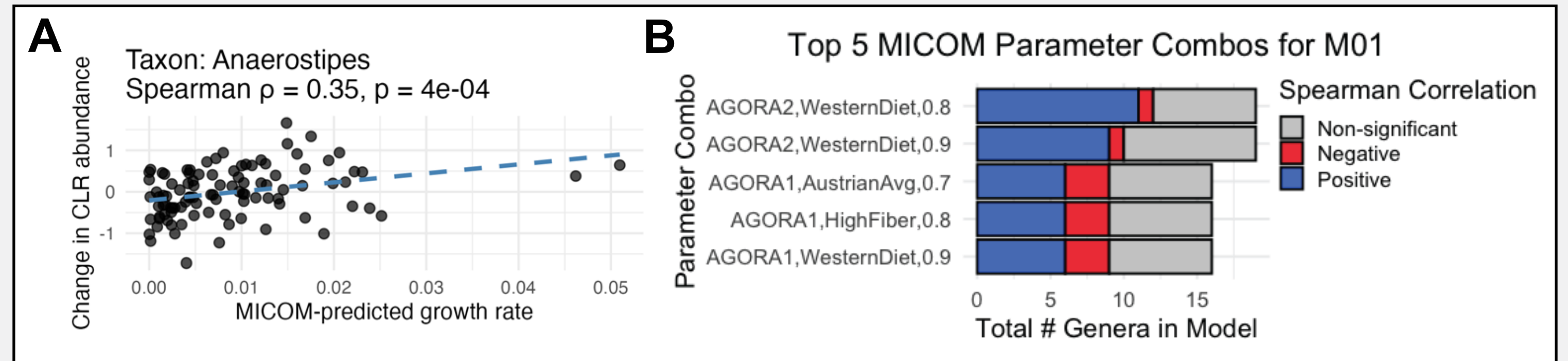


Figure 4. Spearman correlation between observed and MICOM predicted growth rates allows for identification of best-performing parameters. (A) Example significant positive correlation of a single genus (*Anaerostipes*) from a single parameter combination. (B) Top 5 MICOM parameter combinations (GSMM model, diet, tradeoff value) for growth rate prediction.

Genera well-predicted by MICOM serve as initial targets for investigation of TLC interactions

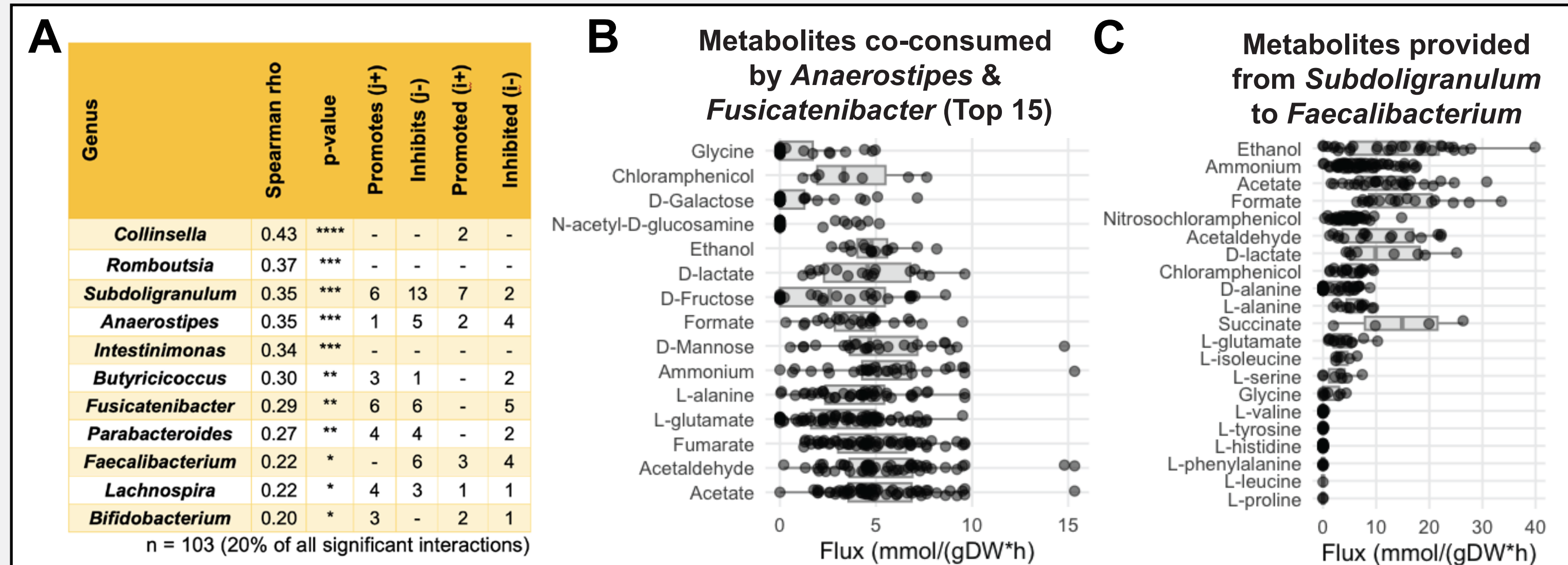


Figure 5. MICOM modeling of significant TLC interactions between genera that were well-predicted by MICOM. (A) List of genera whose growth was predicted in the top MICOM condition and number of times they appeared in TLC significant results. (B-C) Metabolic fluxes underlying (B) a negative TLC interaction (shown by co-consumed metabolites) and (C) a positive TLC interaction (shown as metabolites provided from OTUj to OTUi). Note presence of SCFA acetate, sugars, and amino acids.

Conclusions & Future Directions

- TLC can be used to identify lagged relationships between taxa
- Pairs of microbes with negative TLC relationships are often closely related
- Time series data can be used to choose optimal MICOM parameters
- Ecological relationships inferred from TLC are potentially explained by metabolic relationships as modeled by MICOM
- Identify metabolic fluxes underlying pairwise interactions identified by TLC with MICOM
- Decrease prevalence threshold/add additional filtering metrics to detect less prevalent taxa while retaining reasonable compute time
- Test varying lengths of time lag
- Model before, during, and after disturbance separately
- Test pipeline on other simulated and real datasets

References

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Acknowledgements

This work was supported by the National Library of Medicine (NLM) of the National Institutes of Health (NIH) under award number T15LM009451 (LML). Figures generated in python, RStudio, and Adobe Illustrator.